

Paper 2 - Set 05 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	A	Quadratic functions
2	B	Integration
3	C	Triangles, radians and sectors
4	A	Chains of reasoning
5	A	Chains of reasoning
6	C	The logic of arguments
7	E	The logic of arguments
8	E	Mathematical proof
9	C	Simultaneous equations and inequalities
10	F	Angles, polygons, congruence and vectors
11	C	Identifying errors in proofs
12	E	Trigonometric functions and equations
13	H	Indices and surds
14	A	Mathematical proof
15	C	Mathematical proof
16	C	Differentiation
17	E	Chains of reasoning
18	A	Identifying errors in proofs
19	H	Sequences and series
20	D	Integration

Worked solutions

Question 1 QUA-21 · Quadratic functions · spec MM1.3

The parabola $y = x^2 - 4x + 3$ is reflected in the x -axis and then translated 2 units in the positive y -direction. What is the maximum value of the resulting function?

Answer: A

$x^2 - 4x + 3 = (x - 2)^2 - 1$, minimum -1 . Reflecting gives maximum 1; translating up by 2 gives maximum 3.

Question 2 INT-06 · Integration · spec MM7.5

The trapezium rule with 2 strips of equal width is used to estimate $\int_0^2 x^2 dx$. Which of the following is true?

Answer: B

$h = 1$: $\frac{h}{2}[y_0 + 2y_1 + y_2] = \frac{1}{2}[0 + 2(1) + 4] = 3$. The true value is $\frac{8}{3}$; the curve bends upwards (convex), so the chords lie above it and the rule overestimates.

Question 3 TRI-05 · Triangles, radians and sectors · spec MM4.2

Convert 135° to radians.

Answer: C

$135 = \frac{3}{4} \times 180$, so $\frac{3\pi}{4}$.

Question 4 RSN-10 · Chains of reasoning · spec Prf5

Each of 100 cards has a different integer from 1 to 100 written on it. What is the smallest number of cards that must be drawn (without looking) to be certain of having two cards whose numbers sum to 101?

Answer: A

The 50 pairs $\{1, 100\}, \{2, 99\}, \dots, \{50, 51\}$ each sum to 101. Fifty cards (one from each pair) may avoid it; 51 cannot.

Question 5 RSN-05 · Chains of reasoning · spec Prf5

In a knockout tournament with 37 players, each match eliminates exactly one player and the tournament ends when one player remains. How many matches are played?

Answer: A

Each match eliminates one player and 36 must be eliminated, so 36 matches - whatever the draw looks like.

Question 6 LGC-11 · The logic of arguments · spec Arg1

The statement 'if a triangle is equilateral then it is isosceles' is true. Which of the following must therefore also be true?

- I If a triangle is isosceles then it is equilateral.
- II If a triangle is not isosceles then it is not equilateral.
- III If a triangle is not equilateral then it is not isosceles.

Answer: C

Only the contrapositive (II) is logically equivalent to the original. I is the converse and III is the inverse; both are in fact false here.

Question 7 LGC-13 · The logic of arguments · spec Arg4

Which of the following is the negation of $x > 3$ and $y \leq 5$?

Answer: E

Not $(A \text{ and } B)$ is $(\text{not } A) \text{ or } (\text{not } B)$. Not $x > 3$ is $x \leq 3$; not $y \leq 5$ is $y > 5$.

Question 8 PRF-01 · Mathematical proof · spec Prf1

Consider the claim: 'for every positive integer n , $n^2 + n$ is even.' Which of the following is a complete and valid proof?

Answer: E

A covers every n with a general argument. B checks only three cases; C handles only even n ; D is false for odd n ; E is incomplete.

Question 9 X2-14 · Simultaneous equations and inequalities · spec MM1.5

x is a real number. $P: x^2 < 4$. $Q: x < 2$. Which one of the following is true?

Answer: C

$x^2 < 4$ means $-2 < x < 2$, which implies $x < 2$ (sufficient); but $x = -5$ satisfies Q and not P (not necessary).

Question 10 GEO-10 · Angles, polygons, congruence and vectors · spec M5.3

Which of the following statements is/are true?

- I Every rhombus is a parallelogram.
- II Every rectangle is a rhombus.
- III The diagonals of a kite are perpendicular.

Answer: F

A rhombus has two pairs of parallel sides (I). A rectangle need not have equal sides (II false). The diagonals of a kite cross at right angles (III).

Question 11 ERR-14 · Identifying errors in proofs · spec Err2

Claim: if x and y are real with $x^2 = y^2$, then $x = y$.

Which of the following is a counterexample to the claim?

Answer: C

A counterexample must satisfy the hypothesis ($x^2 = y^2$) and fail the conclusion ($x \neq y$).

Question 12 X2-07 · Trigonometric functions and equations · spec MM4.6

Consider the equation $\sin x = k$ where k is a real constant. Which of the following statements is/are true?

- I For every k with $-1 < k < 1$ the equation has exactly two solutions in $0 \leq x < 2\pi$.
- II For $k = 1$ the equation has exactly one solution in $0 \leq x < 2\pi$.
- III For every real k the equation has at least one solution.

Answer: E

I and II are read from the graph of $\sin$ over one period. III is false: $\sin x = 2$ has no solutions.

Question 13 X2-12 · Indices and surds · spec MM1.1

a and b are positive real numbers. Which of the following statements is/are true?

- I If $a^2 = b^2$ then $a = b$.
- II If $2^a = 2^b$ then $a = b$.
- III If $a^b = 1$ then $a = 1$.

Answer: H

I holds because $a, b > 0$ rules out $a = -b$. II holds because 2^x is one-to-one. III: for positive a and b , $a^b = 1$ gives $b \log a = 0$, so $\log a = 0$ and $a = 1$. All three are true.

Question 14 PRF-11 · Mathematical proof · spec Prf3

A student notices that $1 + 2 = 3$, $4 + 5 + 6 = 7 + 8$, $9 + 10 + 11 + 12 = 13 + 14 + 15$ and conjectures that the pattern continues. Which of the following describes the n th line of the pattern?

Answer: A

Line 1 starts at 1, line 2 at 4, line 3 at 9: start n^2 . Left sides have 2, 3, 4 numbers ($n + 1$), right sides 1, 2, 3 (n).

Question 15 PRF-06 · Mathematical proof · spec Prf1

Which of the following statements can be disproved by a single counterexample?

- I Every prime number greater than 2 is odd.
- II Every integer of the form $4k + 3$ is prime.
- III There exists an even prime number.

Answer: C

Only a 'for all' statement can be disproved by one counterexample. I is true (no counterexample exists). II is false and $k = 3$ (15) disproves it. III is an existence statement - a counterexample is meaningless; it is proved by the example 2.

Question 16 X2-02 · Differentiation · spec MM6.3

Let f be a polynomial. P : ' $f'(x) > 0$ for all real x '. Q : ' $f(x) = 0$ has exactly one real root'. Which one of the following is true?

Answer: C

If $f' > 0$ everywhere, f is strictly increasing and (being an odd-degree polynomial) crosses the axis exactly once: P is sufficient. But $f(x) = x^3$ has exactly one root and $f'(0) = 0$, so P is not necessary.

Question 17 RSN-19 · Chains of reasoning · spec Prf5

A bag contains red and blue balls only. If one red ball is removed, one quarter of the remaining balls are red. If instead two blue balls are removed, one third of the remaining balls are red. How many balls are in the bag?

Answer: E

With r red and b blue: $r - 1 = \frac{1}{4}(r + b - 1)$ and $r = \frac{1}{3}(r + b - 2)$. The second gives $b = 2r + 2$; substituting into the first, $4r - 4 = 3r + 1$, so $r = 5$, $b = 12$: 17 balls.

Question 18 ERR-05 · Identifying errors in proofs · spec Err2

A student solves the inequality $\frac{x+1}{x-2} < 3$.

I Multiply both sides by $(x-2)$: $x+1 < 3x-6$.

II So $7 < 2x$.

III So $x > \frac{7}{2}$.

Which of the following is correct?

Answer: A

For $x < 2$ the multiplier is negative and the inequality reverses; indeed $x = 0$ gives $-\frac{1}{2} < 3$, a solution missed by the student. The full solution set is $x < 2$ or $x > \frac{7}{2}$.

Question 19 SEQ-21 · Sequences and series · spec MM2.1

A sequence has $u_1 = 3$ and $u_{n+1} = \frac{u_n + 5}{2}$. Which of the following statements is/are true?

I The sequence is increasing.

II $u_n < 5$ for all n .

III $u_n = 5 - 2^{2-n}$ for all n .

Answer: H

$u_2 = 4$, $u_3 = 4.5$, $u_4 = 4.75$. If $u_n < 5$ then $u_{n+1} = \frac{u_n + 5}{2} < 5$ and $u_{n+1} - u_n = \frac{5 - u_n}{2} > 0$, so I and II hold. III: $5 - 2^{2-n}$ gives 3, 4, 4.5, ... and satisfies the recurrence: $\frac{5 - 2^{2-n} + 5}{2} = 5 - 2^{1-n}$. All three are true.

Question 20 X2-03 · Integration · spec MM7.1

f is a function defined for all real x and $\int_0^4 f(x) dx = 0$. Which of the following statements must be true?

- I $f(x) = 0$ for some x with $0 \leq x \leq 4$.
- II The area between the curve and the x -axis from 0 to 4 is 0.
- III $\int_0^2 f(x) dx = -\int_2^4 f(x) dx$

Answer: D

III is just the rule for combining integrals over adjacent ranges. I fails for a function that jumps from -1 to 1 without passing through 0 (nothing says f is continuous). II fails: $f(x) = x - 2$ has integral 0 but area 4.